

N° 8203

PSEUDO MAXIMUM LIKELIHOOD METHODS :
APPLICATIONS TO POISSON MODELS

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1. INTRODUCTION.

The analysis of economic behaviors often leads to the study of characteristics taking a small number of positive values. The classical linear model is not adapted to explain how such discrete variables depend on other quantitative or qualitative variables. The reasons are similar to those usually given in the case of an endogenous qualitative variable : the shape of the observations set does not correspond to a linear model ; the assumption of normality of the disturbances cannot be made, since the endogenous variable takes a small number of values with strictly positive probabilities ; the formulae of prediction which are deduced from linear model give impossible values...

In the models considered in the literature to describe discrete variables (El. SAYYAD (1973), GILBERT (1979), HAUSMAN, HALL, GRILICHES (1981)) the endogenous variable is assumed to have a Poisson distribution conditionally to the exogenous variables. The parameter of this distribution is a function of the values of the exogenous variables.

The choice of such a model is justified if the dependent variable counts the occurrence of a given event during a fixed period and if the usual assumptions of Poisson process are satisfied. For instance, the model is adapted to describe daily numbers of oil tanker's arrivals in a port, the number of accidents at work by factory, the number of patents

applied for and received by firms (HAUSMAN, HALL, GRILICHES (1981))...

This kind of models may also be used as a first approach for cases where the usual assumptions are not exactly satisfied in particular the assumption of independence between the present and the past. For instance, such a model has been used by GILBERT (1979) to explain the number of jobs held during a year.

In section 2 we give the definitions of the basic Poisson model and of the Poisson model with specification errors. In the later case, we show how the first and second order moments of the endogenous variable depends on the parameters (section 3), and we derive pseudo maximum likelihood estimation methods only based on these first and second moments (GOURIEROUX, MONFORT, TROGNON (1981)). Generalizations of Poisson model for the multivariate case, in particular for models required by panel data are described in section 4 and 5.

2. PRELIMINARIES.

In this section, the basic Poisson model and the Poisson model with specification errors are described.

2.a. The basic Poisson model.

Let y_i $i = 1, \dots, n$ be the observations of a discrete variable. In the basic Poisson model, it is assumed that the y_i are

independently distributed and that the distribution of y_i is a Poisson distribution, whose parameter is

$$\lambda_i = \exp(x_i b) = \exp \left(\sum_{k=1}^K x_{ik} b_k \right)$$

where $(x_{i1}, \dots, x_{iK})$ are K exogenous variables associated with the i^{th} observation and $(b_1, \dots, b_K)$ are K unknown parameters.

The exponential function appearing in λ_i is mainly justified by the positivity of λ_i ; a linear function $\lambda_i = x_i b$ would have implied possibly incompatible constraints on the parameters. The mean and the variance of y_i are equal to λ_i and the density function of y_i is :

$$\frac{\exp(-\lambda_i) \lambda_i^{y_i}}{y_i!}$$

The log-likelihood function is :

$$\begin{aligned} L(b) &= - \sum_{i=1}^n \lambda_i + \sum_{i=1}^n y_i \text{Log } \lambda_i - \sum_{i=1}^n \text{Log } (y_i!) \\ &= \text{constant} - \sum_{i=1}^n \exp(x_i b) + \sum_{i=1}^n y_i x_i b \end{aligned}$$

In the maximization of $L(b)$, the first order conditions are :

$$\frac{dL}{db} = - \sum_{i=1}^n x_i [\exp(x_i b) - y_i] = 0$$

$$\text{or} \quad \sum_{i=1}^n x_i (\lambda_i - y_i) = 0^{(1)}$$

If a solution of the previous equations exists⁽²⁾, it is a unique maximum since the Hessian matrix $\frac{d^2L}{db db'} = - \sum_{i=1}^n x_i' x_i \exp(x_i' b)$ is negative definite, if $X'X = \sum_{i=1}^n x_i' x_i$ is of full rank; the log-likelihood function is, in this case, strictly concave and can be easily maximized by the usual algorithms.

2.b. The Poisson model with specification error.

In the previous model the conditional expectation of the endogenous variable given the exogenous variables and the corresponding conditional variance cannot vary independently⁽³⁾. To overcome this restriction it has been proposed to introduce a disturbance in the definition of the parameter of the Poisson distribution. In such models the parameter is written :

$$\lambda_i^* = \exp(x_i' b + \varepsilon_i)$$

where ε_i is a specification error due, for instance, to omitted explanatory variables independent from the exogenous variables x . This randomness of the model must be distinguished from that of the Poisson model, which is related to the random character of the endogenous variable considered. In this model the λ_i^* are random variables, and the conditional distribution of the y_i , $i = 1, \dots, n$, given the x_i and ε_i (or λ_i^*) $i = 1, \dots, n$ is given by

$$\prod_{i=1}^n \ell(y_i | x_i, \varepsilon_i), \text{ where } \ell(y_i | x_i, \varepsilon_i) \text{ is the density of}$$

the Poisson distribution $P[\exp(x_i' b + \varepsilon)]$.

Since ϵ_i is an unobservable random variable, we must integrate it out to obtain the conditional distribution of y_i given x_i :

$$\begin{aligned} \ell^*(y_i|x_i) &= \int \frac{\exp(-\lambda_i^*) \lambda_i^{*y_i}}{y_i!} g(\epsilon_i) d\epsilon_i \\ &= \int \frac{\exp[-\exp(x_i b + \epsilon_i)] \exp[y_i(x_i b + \epsilon_i)]}{y_i!} g(\epsilon_i) d\epsilon_i \end{aligned}$$

where g is the p.d.f of ϵ_i .

If the disturbances ϵ_i $i = 1, \dots, N$ are i.i.d. the log-likelihood function is given by :

$$L^* = \sum_{i=1}^n \text{Log } \ell^*(y_i|x_i)$$

The distribution ℓ^* depends on the distribution of ϵ_i and, in general, ℓ^* is a very complicated function. For convenience, it is sometimes assumed that $u_i = \exp(\epsilon_i)$ is distributed as a $\gamma(\alpha, \beta)$, with p.d.f :

$$\frac{u^{\alpha-1} \exp(-u/\beta)}{\beta^\alpha \Gamma(\alpha)}$$

This kind of models has been considered by HAUSMAN, HALL, GRILICHES (1981) and by GILBERT (1979).

Assuming that there exists a constant term in $x_i b$, there is no loss of generality to set : $E \exp(\epsilon_i) = 1$. In that case $\beta = \alpha^{-1} = \eta^2 = V[\exp(\epsilon_i)]$ and it is straightforward to show (see for

instance GILBERT (1979)) that the distribution of y_i given x_i is a negative binomial distribution :

$$\ell^*(y_i|x_i; b, \eta^2) = \frac{\Gamma(\frac{1}{\eta^2} + y_i)}{\Gamma(\frac{1}{\eta^2})\Gamma(y_i+1)} \left[\eta^2 \exp(x_i b) \right]^{y_i} \left[1 + \eta^2 \exp(x_i b) \right]^{-\left(\frac{1}{\eta^2} + y_i\right)}$$

The ML estimation of b can be computed by numerical methods which necessitate expansions of gamma and digamma functions.

As shown later, this ML estimator may be inconsistent if $\exp(\epsilon)$ does not have a gamma distribution. Therefore it is interesting to propose simple estimation methods leading to consistent estimators for all possible distribution g having second order moments.

3. ESTIMATION METHOD FOR THE POISSON MODEL WITH SPECIFICATION ERROR.

Since $E(\exp \epsilon_i) = 1$ and $V(\exp \epsilon_i) = \eta^2$, we have :

$$\begin{aligned} E(y_i|x_i) &= E E(y_i|\epsilon_i, x_i) = E \exp(x_i b + \epsilon_i) \\ &= \exp(x_i b) E \exp(\epsilon_i) \\ &= \exp(x_i b) \end{aligned}$$

$$\begin{aligned} V(y_i|x_i) &= E(V(y_i|x_i, \epsilon_i)) + V(E(y_i|x_i, \epsilon_i)) \\ &= E(\exp(x_i b + \epsilon_i)) + V(\exp(x_i b + \epsilon_i)) \\ &= \exp(x_i b) + \eta^2 \exp(2x_i b) . \end{aligned}$$

These results will be used to estimate the true values b_0, η_0^2 of b, η^2 without assumptions on g .

We shall consider three kinds of methods proposed in GMT (1981), which are Pseudo Maximum Likelihood of type I (PMLI), Quasi-Generalised Pseudo Maximum Likelihood (QGPMML) and Pseudo maximum likelihood of type II (PML II).

3.a. Pseudo Maximum Likelihood Estimator of type I.

We shall consider the PML estimator associated to four exponential families of type I : the normal family (with unit variance), the Poisson family, the Negative Binomial family (with a η^2 taken equal to a given positive number a) and the gamma family.

The objective functions corresponding to these families are respectively :

$$(i) \quad - \sum_{i=1}^n (y_i - \exp x_i b)^2 \quad (\text{nonlinear least squares})$$

$$(ii) \quad - \sum_{i=1}^n \exp x_i b + \sum_{i=1}^n y_i x_i b$$

$$(iii) \quad \sum_{i=1}^n \left\{ y_i x_i b - \left(\frac{1}{a} + y_i \right) \text{Log} (1 + a \exp x_i b) \right\}$$

$$(iv) \quad - \sum_{i=1}^n \left[- x_i b - y_i \exp(- x_i b) \right]$$

The pseudo-likelihood equations are :

$$(i) \quad -2 \sum_{i=1}^n x_i (y_i - \exp x_i b) \exp x_i b = 0$$

$$(ii) \quad \sum_{i=1}^n x_i (-\exp x_i b + y_i) = 0$$

$$(iii) \quad \sum_{i=1}^n x_i \frac{y_i - \exp x_i b}{1 + a \exp x_i b} = 0$$

$$(iv) \quad \sum_{i=1}^n x_i (y_i - \exp x_i b) \exp (-x_i b) = 0$$

The Hessian matrices of the objective functions are :

$$(i) \quad -2 \sum_{i=1}^n x_i' \cdot x_i (y_i - 2 \exp x_i b) \exp x_i b$$

$$(ii) \quad - \sum_{i=1}^n x_i' \cdot x_i \exp x_i b$$

$$(iii) \quad - \sum_{i=1}^n \frac{x_i' \cdot x_i (1 + 2a \exp x_i b) \exp x_i b}{(1 + a \exp x_i b)^2}$$

$$(iv) \quad - \sum_{i=1}^n x_i' \cdot x_i y_i \exp (-x_i b)$$

Note that the last three objective functions are concave and can be easily maximized by usual iterative methods, whereas the objective function associated to the nonlinear least squares is not concave.

We know from GMT (1981) theorems 2 and 3, that these four PML estimators $\hat{b}_j$ $j = 1, \dots, 4$ are strongly consistent and asymptotically normal and that the asymptotic covariance matrix of $\sqrt{n}(\hat{b}_j - b_0)$ has the general form $J^{-1} I J^{-1}$, where

$$J = E_x \left(\frac{\partial f}{\partial b} \Sigma^{-1} \frac{\partial f'}{\partial b} \right)$$

$$I = E_x \left(\frac{\partial f}{\partial b} \Sigma^{-1} \Sigma_0 \Sigma^{-1} \frac{\partial f'}{\partial b} \right)$$

with

$$f(b) = \exp x' b$$

$$\frac{\partial f}{\partial b} = x' \exp x' b$$

Σ : covariance matrix of the pseudo distribution

Σ_0 : covariance matrix of the true distribution

$\frac{\partial f}{\partial b}$ and Σ being evaluated in b_0 , and Σ_0 being evaluated in (b_0, η_0^2)

The asymptotic covariance matrices are

$$(i) \quad \left[E_x(x' x \exp 2x' b_0) \right]^{-1} E_x \left\{ x' x \exp 2x' b_0 (\exp x' b_0 + \eta_0^2 \exp 2x' b_0) \right\} \left[E_x(x' x \exp 2x' b_0) \right]^{-1}$$

This matrix is consistently estimated by

$$n \left(\sum_i x_i' x_i \exp 2x_i' \hat{b}_n \right)^{-1} \left(\sum_{i=1}^n x_i' x_i \exp 2x_i' \hat{b}_n (y_i - \exp x_i' \hat{b}_n)^2 \right) \left(\sum_{i=1}^n x_i' x_i \exp 2x_i' \hat{b}_n \right)^{-1}$$

$$(ii) \quad \left[E_x(x' x \exp x' b_0) \right]^{-1} E_x \left\{ x' x (\exp x' b_0 + \eta_0^2 \exp 2x' b_0) \right\} \left[E_x(x' x \exp x' b_0) \right]^{-1} \\ = \left[E_x(x' x \exp x' b_0) \right]^{-1} + \eta_0^2 \left[E_x(x' x \exp x' b_0) \right]^{-1} E_x(x' x \exp 2x' b_0) \left[E_x(x' x \exp x' b_0) \right]^{-1}$$

Note that, if the likelihood of the basic Poisson model is used, the estimator obtained is consistent and asymptotically normal ; however the previous formula shows that the asymptotic covariance matrix obtained

when it is (wrongly) assumed that the correct model is the basic Poisson model, i.e. $\left[E_x(x'x \exp x b_0) \right]^{-1}$, is "smaller" than the true one

$$(iii) \left[E_x \left(\frac{x'x \exp x b_0}{1 + a \exp x b_0} \right) \right]^{-1} E_x \left\{ \frac{x'x (\exp x b_0 + \eta_0^2 \exp 2x b_0)}{(1 + a \exp x b_0)^2} \right\} \left[E_x \left(\frac{x'x \exp x b_0}{1 + a \exp x b_0} \right) \right]^{-1}$$

$$(iv) \left(E_x x'x \right)^{-1} E_x \left\{ x'x [\eta_0^2 + \exp(-x b_0)] \right\} \left(E_x x'x \right)^{-1} \\ = \eta_0^2 \left(E_x x'x \right)^{-1} + \left(E_x x'x \right)^{-1} E_x [x'x \exp(-x b_0)] \left(E_x x'x \right)^{-1}$$

It might have seemed natural to consider a generalized least square estimator of b by minimizing the following objective function

$$\sum_{i=1}^n \frac{(y_i - \exp(x_i b))^2}{\exp(x_i b) + a \exp(2x_i b)}$$

where a is a given number.

However such a method based on the criterion :

$$\varphi(u, m) = \frac{(u-m)^2}{m + am^2}$$

is not always consistent. This result is a consequence of the proof of theorem 2 in GMT (1981), which establishes a necessary condition for the consistency of the estimator obtained by minimizing $\sum_i \varphi(y_i, \exp(x_i b))$. This condition is that $\frac{\partial \varphi}{\partial m} = \lambda(m)(u-m)$ which is not satisfied in the present case.

3.b. Quasi-generalized pseudo maximum likelihood estimators.

To apply the QGPML approach (GMT (1981) section 5), it is necessary to have consistent estimators of b and η^2 . b can be consistently estimated by any $\hat{b}_j$ and a consistent estimator of η^2 may be derived along the following lines : since $\forall y_i = \exp(x_i b) + \eta^2 \exp(2x_i b)$, we have : $(y_i - \exp(x_i b))^2 - \exp(x_i b) = \eta^2 \exp(2x_i b) + u_i$ with $E u_i = 0$. A strongly consistent estimator of η^2 is the estimated regression coefficient :

$$\hat{\eta}_j^2 = \frac{\sum_{i=1}^n [(y_i - \exp(x_i \hat{b}_j))^2 - \exp(x_i \hat{b}_j)] \exp(2x_i \hat{b}_j)}{\sum_{i=1}^n \exp(4x_i \hat{b}_j)}$$

QGPML estimators can be obtained by using the normal family (i), the negative binomial family (iii) and the gamma family (iv). In these cases the objective functions are :

$$(i) \quad \sum_{i=1}^n \frac{(y_i - \exp(x_i b))^2}{\exp(x_i \hat{b}_1) + \hat{\eta}_1^2 \exp(2x_i \hat{b}_1)}$$

$$(iii) \quad \sum_{i=1}^n y_i x_i b - \left(\frac{1}{\hat{\eta}_3^2} + y_i \right) \text{Log}(1 + \hat{\eta}_3^2 \exp(x_i b))$$

$$(iv) \quad \sum_{i=1}^n \left\{ \frac{\exp(x_i \hat{b}_4)}{1 + \hat{\eta}_4^2 \exp(x_i \hat{b}_4)} (-x_i b - y_i \exp(-x_i b)) \right\}$$

As previously mentioned, the two last objective functions are globally concave (if $\hat{\eta}_j^2 \leq 0$, we take $\hat{\eta}_j^2 = 0$ and if $\hat{\eta}_3^2 = 0$, (iii) should be replaced by the function associated to the basic Poisson model).

All these estimators are consistent and have the same asymptotic covariance matrix : $\left[E_x \left(\frac{x'x \exp(xb_0)}{1 + \eta_0^2 \exp(xb_0)} \right) \right]^{-1}$. This covariance matrix is smaller than the asymptotic covariance matrices of the PML estimators of the previous section.

If we expand the covariance matrix of the QGPML estimators in a neighbourhood of $\eta_0^2 = 0$, we have :

$$\begin{aligned} \left[E_x \left(\frac{x'x \exp(xb_0)}{1 + \eta_0^2 \exp(xb_0)} \right) \right]^{-1} &= \left[E_x \left(x'x \exp(xb_0) (1 - \eta_0^2 \exp(xb_0)) \right) \right]^{-1} + o(\eta_0^2) \\ &= \left[E_x (x'x \exp(xb_0)) - \eta_0^2 E_x (x'x \exp(2xb_0)) \right]^{-1} + o(\eta_0^2) \\ &= \left[E_x (x'x \exp(xb_0)) \right]^{-1} + \eta_0^2 \left[E_x (x'x \exp(xb_0)) \right]^{-1} \\ &\quad \left[E_x (x'x \exp(2xb_0)) \right] \left[E_x (x'x \exp(xb_0)) \right]^{-1} + o(\eta_0^2) \end{aligned}$$

This shows that the PML estimator based on the Poisson family is second order efficient in a neighbourhood of $\eta_0^2 = 0$.

3.c. Pseudo maximum likelihood estimator of type II.

The parameters b and η^2 can be estimated simultaneously by using the PML II approach. This method may be based for instance on the normal family. The estimators are defined as the solutions of the optimisation problem :

$$\text{Max}_{b, \eta^2} \Psi(b, \eta^2) = -\frac{1}{2} \sum_{i=1}^n \text{Log}[\exp(x_i b) + \eta^2 \exp(2x_i b)] - \frac{1}{2} \sum_{i=1}^n \frac{[y_i - \exp(x_i b)]^2}{\exp(x_i b) + \eta^2 \exp(2x_i b)}$$

$$\begin{aligned}\frac{\partial \Psi(b, \eta^2)}{\partial b} &= \sum_{i=1}^n x_i \left\{ \frac{1+2\eta^2 \exp(x_i b)}{1+\eta^2 \exp(x_i b)} + \frac{[y_i - \exp(x_i b)]^2 [1+2\eta^2 \exp(x_i b)]}{[1+\eta^2 \exp(x_i b)]^2 \exp(x_i b)} \right. \\ &\quad \left. - 2 \frac{y_i - \exp(x_i b)}{1+\eta^2 \exp(x_i b)} \right\} = 0 \\ \frac{\partial \Psi(b, \eta^2)}{\partial \eta^2} &= \sum_{i=1}^n \left\{ \frac{\exp(x_i b)}{1+\eta^2 \exp(x_i b)} - \frac{[y_i - \exp(x_i b)]^2}{[1+\eta^2 \exp(x_i b)]^2} \right\} = 0\end{aligned}$$

The asymptotic covariance matrix of the PMLII estimators is obtained by applying the result of (GMT (1981) : appendix 4), with $C(m, \sigma^2) = \frac{m}{\sigma^2}$, $D(m, \sigma^2) = -\frac{1}{2\sigma^2}$

$$h(b, \eta^2) = \begin{bmatrix} \exp xb \\ \exp xb + \eta^2 \exp 2xb \end{bmatrix}$$

$$\frac{\partial h(b, \eta^2)}{\partial (b, \eta^2)} = \begin{bmatrix} x' \exp(xb) & x' [\exp(xb) + 2\eta^2 \exp(2xb)] \\ 0 & \exp(2xb) \end{bmatrix}$$

Finally, we can remark that the objective function associated with the negative binomial family, i.e.

$$\sum_{i=1}^n \text{Log} \left[\frac{\Gamma(\frac{1}{\eta^2} + y_i)}{\Gamma(\frac{1}{\eta^2}) \Gamma(y_i + 1)} [\eta^2 \exp(x_i b)]^{y_i} [1 + \eta^2 \exp(x_i b)]^{-\left(\frac{1}{\eta^2} + y_i\right)} \right]$$

does not satisfy the necessary condition of theorem 7 in GMT (1981) for the strong consistency of the PMLII estimator. Therefore the method described in 2.b. is inconsistent for some true distributions of $\exp \varepsilon_i$ (different from the gamma distribution).

4. THE BIVARIATE POISSON MODEL.

4.a. The statistical model.

In this section it will be proposed a generalization of Poisson models which allows to describe simultaneously count variables y_1 and y_2 . The corresponding bivariate Poisson model is the same as the model presented by JOHNSON and KOTZ (1969) p. 297-300. This model can be obtained through classical hypotheses of Poisson processes which are the following.

H1 : $y_1(t)$ and $y_2(t)$ are the number of events of types 1 and 2 which occur between 0 and t .

H2 : the occurrence of some events between t and $t + dt$ is independent of what occurred before t .

H3 : between t and $t + dt$, it will occur :

i) one event of type 1 and no event of type 2 with probability

$$\lambda dt + o(dt) \left[\lim_{dt \rightarrow 0} \frac{o(dt)}{dt} = 0 \right]$$

ii) one event of type 2 and no event of type 1 with probability

$$\mu dt + o(dt)$$

iii) one event of type 1 and one event of type 2 with probability

$$\gamma dt + o(dt)$$

iv) no event with probability $1 - \lambda dt - \mu dt - \gamma dt + o(dt)$

The distribution of the process $[y_1(t), y_2(t)]$ can be

easily obtained if we denote by $P_{n,m}(t)$ the probability that $y_1(t) = n$ and $y_2(t) = m$ and by $G(t,u,v) = E u^{y_1(t)} v^{y_2(t)}$ the moment generating function. By definition, we have the following difference equation

$$\begin{aligned} P_{n,m}(t+dt) = & P_{n,m}(t) [1 - \lambda dt - \mu dt - \gamma dt + o(dt)] \\ & + P_{n-1,m}(t) [\lambda dt + o(dt)] + P_{n,m-1}(t) [\mu dt + o(dt)] \\ & + P_{n-1,m-1}(t) [\gamma dt + o(dt)] \end{aligned}$$

As dt tends to zero, we can write :

$$\frac{dP_{n,m}(t)}{dt} = (-\lambda - \mu - \gamma)P_{n,m}(t) + \lambda P_{n-1,m}(t) + \mu P_{n,m-1}(t) + \gamma P_{n-1,m-1}(t)$$

This relation implies the following differential equation for the moment generating function :

$$\frac{\partial}{\partial t} G(t,u,v) = (-\lambda - \mu - \gamma + \lambda u + \mu v + \gamma uv) G(t,u,v)$$

$$\text{Then : } G(t,u,v) = \exp\{(-\lambda - \mu - \gamma + \lambda u + \mu v + \gamma uv)t\}$$

The expression of G involves that $y_1(t)$ and $y_2(t)$ are respectively Poisson variables with parameters $(\lambda + \gamma)t$ and $(\mu + \gamma)t$ setting $u=1$ or $v=1$. These variables are correlated. The covariance between $y_1(t)$ and $y_2(t)$ is obtained through the second cross partial derivative of G :

$$E y_1(t)y_2(t) = \left[\frac{\partial^2 G(t,u,v)}{\partial u \partial v} \right]_{u=1,v=1} = \gamma t + (\lambda + \gamma)t \cdot (\mu + \gamma)t$$

$$\text{and } \text{Cov}(y_1(t), y_2(t)) = E y_1(t)y_2(t) - E y_1(t) \cdot E y_2(t) = \gamma t \quad (4)$$

Note that this model can be obtained by setting :

$$y_1(t) = \alpha(t) + \beta_1(t)$$

$$y_2(t) = \alpha(t) + \beta_2(t)$$

where $\alpha(t), \beta_1(t), \beta_2(t)$ are independent Poisson variables with parameters $\gamma t, \lambda t$ and μt respectively. This remark will be used in the multivariate case (see section 5). At this stage, it is possible to note that the difference $y_1(t) - y_2(t)$ is equal to $\beta_1(t) - \beta_2(t)$; this means that, if only $y_1(t) - y_2(t)$ is observable, γ is not identifiable and the independence between $y_1(t)$ and $y_2(t)$ can be assumed. More importantly, the difference between two independent Poisson variables provides a natural generalization of the usual Poisson variable which can be used when the discrete variable considered can take negative values. The specifications and the estimation methods proposed above can be easily extended to this case

4.b. The econometric model.

Let us consider a sample of n independent observations of (y_{1i}, y_{2i}) distributed as bivariate Poisson variables ($i=1,2,\dots,n$). These observations are counts of events of types 1 and 2 during a period of length 1 (by convention). The econometric model will be complete if we define the moments of (y_{1i}, y_{2i}) as functions of explanatory variables and disturbances; by analogy with the previous section we will assume :

$$\lambda_i = \exp(x_i b + \varepsilon_{1i}) \quad , \quad \mu_i = \exp(z_i c + \varepsilon_{2i})$$

$$\gamma_i = \exp(m_i d + \varepsilon_{3i})$$

with

$$E \exp \varepsilon_{1i} = E \exp \varepsilon_{2i} = E \exp \varepsilon_{3i} = 1$$

$$V \begin{bmatrix} \exp \epsilon_{1i} \\ \exp \epsilon_{2i} \\ \exp \epsilon_{3i} \end{bmatrix} = \Omega$$

The complexity of the elementary probabilities $P_{n,m}$ does not allow to integrate out the distribution of the disturbances $(\epsilon_{1i}, \epsilon_{2i}, \epsilon_{3i})$. In this bivariate case the maximum likelihood method is not tractable, whereas the estimation methods based on first and second order moments are applicable. The expectations of the observed variables are :

$$E y_{1i} = \exp(x_i b) + \exp(m_i d) \quad , \quad E y_{2i} = \exp(z_i c) + \exp(m_i d)$$

And, the covariance matrix is given by :

$$V \begin{bmatrix} y_{1i} \\ y_{2i} \end{bmatrix} = \begin{bmatrix} \exp(x_i b) + \exp(m_i d) & \exp(m_i d) \\ \exp(m_i d) & \exp(z_i c) + \exp(m_i d) \end{bmatrix} + \begin{bmatrix} \exp(x_i b) & 0 & \exp(m_i d) \\ 0 & \exp(z_i c) & \exp(m_i d) \end{bmatrix} \Omega \begin{bmatrix} \exp(x_i b) & 0 \\ 0 & \exp(z_i c) \\ \exp(m_i d) & \exp(m_i d) \end{bmatrix}$$

The nonlinear least squares estimators of b , c and d are obtained by minimizing :

$$\sum_{i=1}^n \left[(y_{1i} - \exp(x_i b) - \exp(m_i d))^2 + (y_{2i} - \exp(z_i c) - \exp(m_i d))^2 \right]$$

Let us denote by $\hat{b}$, $\hat{c}$, $\hat{d}$ these estimators and by $\hat{w}_{1i}$ and $\hat{w}_{2i}$

the estimated residuals :

$$\begin{aligned}\hat{w}_{1i} &= y_{1i} - \exp(x_i \hat{b}) - \exp(m_i \hat{d}) \\ \hat{w}_{2i} &= y_{2i} - \exp(z_i \hat{c}) - \exp(m_i \hat{d})\end{aligned}$$

Consistent estimators of ω_{ij} $i, j = 1, 2, 3$, the elements of Ω , are obtained by applying ordinary least squares to :

$$\left\{ \begin{aligned}\hat{w}_{1i}^2 - \exp(x_i \hat{b}) - \exp(m_i \hat{d}) &= \omega_{11} \exp(2x_i \hat{b}) + 2\omega_{13} \exp(x_i \hat{b} + m_i \hat{d}) + \omega_{33} \exp(2m_i \hat{d}) \\ &\quad + \text{disturbance} \\ \hat{w}_{2i}^2 - \exp(z_i \hat{c}) - \exp(m_i \hat{d}) &= \omega_{22} \exp(2z_i \hat{c}) + 2\omega_{23} \exp(z_i \hat{c} + m_i \hat{d}) + \omega_{33} \exp(2m_i \hat{d}) \\ &\quad + \text{disturbance} \\ \hat{w}_{1i} \hat{w}_{2i} - \exp(m_i \hat{d}) &= \omega_{12} \exp(x_i \hat{b} + z_i \hat{c}) + \omega_{23} \exp(z_i \hat{c} + m_i \hat{d}) + \omega_{13} \exp(x_i \hat{b} + m_i \hat{d}) \\ &\quad + \omega_{33} \exp(2m_i \hat{d}) + \text{disturbance}\end{aligned}\right.$$

The estimator $\hat{\Omega}$ of Ω may be used to apply the QGPML method. The QGPML estimators of b , c , d are solutions of :

$$\text{Min} \sum_{i=1}^n (y_{1i} - E y_{1i}, y_{2i} - E y_{2i}) \hat{V} \begin{pmatrix} y_{1i} \\ y_{2i} \end{pmatrix}^{-1} \begin{pmatrix} y_{1i} - E y_{1i} \\ y_{2i} - E y_{2i} \end{pmatrix}$$

where $\hat{V} \begin{pmatrix} y_{1i} \\ y_{2i} \end{pmatrix}$ is obtained from $V \begin{pmatrix} y_{1i} \\ y_{2i} \end{pmatrix}$ in which b , c , d , Ω are replaced by $\hat{b}$, $\hat{c}$, $\hat{d}$, $\hat{\Omega}$.

5. THE MULTIVARIATE CASE.

5.a. A statistical model for panel data.

The bivariate Poisson model can be directly generalized for the case of m count variables (see JOHNSON-KOTZ (1969)). In this section, we only present a generalization related to the problem of panel data.

The observed variables are $y_{ij}(t)$ $i = 1, \dots, I$, $j = 1, \dots, J$. We assume that :

$$y_{ij}(t) = \alpha_i(t) + \beta_j(t) + \gamma_{ij}(t) \quad i = 1, \dots, I \quad , \quad j = 1, \dots, J$$

where $\alpha_i(t)$, $\beta_j(t)$ $\gamma_{ij}(t)$ are independent Poisson variables with parameters $\lambda_i t$, $\mu_j t$, $\gamma_{ij} t$.

The previous formulation is an error component model for count data: $\alpha_i(t)$ and $\beta_j(t)$ are the specific effects , $\gamma_{ij}(t)$ is the residual effects.

The moment generating function of $y_{ij}(t)$ $i = 1, \dots, I$, $j = 1, \dots, J$ is $G(t, u_{ij} \quad i = 1, \dots, I \quad , \quad j = 1, \dots, J)$

$$\begin{aligned} &= E \left(\prod_i \prod_j u_{ij}^{y_{ij}(t)} \right) \\ &= E \left(\prod_i \prod_j u_{ij}^{\alpha_i(t) + \beta_j(t) + \gamma_{ij}(t)} \right) \\ &= \prod_i E \left(\left(\prod_j u_{ij} \right)^{\alpha_i(t)} \right) \prod_j E \left(\left(\prod_i u_{ij} \right)^{\beta_j(t)} \right) \prod_i \prod_j E(u_{ij}^{\gamma_{ij}(t)}) \end{aligned}$$

$$= \exp t \left\{ - \sum_i \lambda_i - \sum_j \mu_j - \sum_{ij} \gamma_{ij} + \sum_i \lambda_i (\pi u_{ij}) + \sum_j \mu_j (\pi u_{ij}) + \sum_{ij} \gamma_{ij} u_{ij} \right\} .$$

The marginal distribution of y_{ij} is $P[(\lambda_i + \mu_j + \gamma_{ij})t]$ and the second order moments are given by :

$$\begin{aligned} V y_{ij} &= (\lambda_i + \mu_j + \gamma_{ij})t \\ \text{Cov}(y_{ij}, y_{k\ell}) &= 0 && \text{if } i \neq k, j \neq \ell \\ \text{Cov}(y_{ij}, y_{\ell j}) &= \mu_j t && \text{if } i \neq \ell \\ \text{Cov}(y_{ij}, y_{i\ell}) &= \lambda_i t && \text{if } j \neq \ell \end{aligned}$$

5.b. The associated econometric model.

As previously stated the exogenous variables and the disturbances are introduced in the parameters λ_i , μ_j , γ_{ij} . This introduction is done in such a way that the interpretation of λ_i , μ_j , γ_{ij} as main and specific effects is maintained.

$$\lambda_i = \exp(x_i b + \epsilon_{1ij}), \quad \mu_j = \exp(z_j c + \epsilon_{2ij}),$$

$\gamma_{ij} = \exp(m_{ij} d + \epsilon_{3ij})$, where the disturbances $\epsilon_{1ij}, \epsilon_{2ij}, \epsilon_{3ij}$ are assumed independent, such that :

$$E(\exp \epsilon_{kij}) = 1, \quad V(\exp \epsilon_{kij}) = \omega_k^2 \quad k = 1, 2, 3$$

Integrating with respect to the ϵ 's, and putting by convention $t = 1$ the first and second order moments of y_{ij} are

$$E y_{ij} = \exp(x_i b) + \exp(z_j c) + \exp(m_{ij} d)$$

$$\begin{aligned} V y_{ij} &= E V(y_{ij}/\epsilon) + V E(y_{ij}/\epsilon) \\ &= \exp x_i b + \omega_1^2 \exp 2x_i b + \exp z_j c + \omega_2^2 \exp 2z_j c \\ &\quad + \exp m_{ij} d + \omega_3^2 \exp 2m_{ij} d \end{aligned}$$

$$\text{Cov}(y_{ij}, y_{k\ell}) = 0 \quad \text{if } i \neq k, j \neq \ell$$

$$\text{Cov}(y_{ij}, y_{\ell j}) = \exp(z_j c) + \omega_2^2 \exp(2z_j c) \quad \text{if } i \neq \ell$$

$$\text{Cov}(y_{ij}, y_{i\ell}) = \exp(x_i b) + \omega_1^2 \exp(2x_i b) \quad \text{if } j \neq \ell$$

The previous approach of double indexed count data provides an alternative formulation to that proposed by HAUSMAN-HALL-GRILICHES (1981).

The parameters may be estimated by QGPML method.

6. CONCLUDING REMARKS.

The properties of the Poisson model presented in this paper are similar to those of gaussian linear models : the distribution of the observations is characterized by its first and second order moments and, in the multidimensional case, the lack of correlation between two endogenous variables is equivalent to the independence between these

variables. These models are also extended in order to define, in the count data context, the analogue of seemingly correlated equations models and errors components models. Simple estimation methods based on the general frameworks proposed in GMT (1981) are available for all these models. It would also be interesting to propose Poisson models where the occurrence of some events between t and $t + dt$ depends of that occurred before t ; such models would be similar to autoregressive model and would be of particular importance for instance for the problems related to job search (See HECKMAN (1981)).

FOOTNOTES

- (1) If a component of x_i is constant the corresponding likelihood equation is :

$$\sum_{i=1}^n (\lambda_i - y_i) = 0$$

This equation implies that the sum of the residuals is equal to zero ; therefore the comparison of the computed sum of the residuals with zero can be used as an indicator of the algorithm accuracy.

- (2) These equations are not always solvable. Let us consider the case $K = 1$; the likelihood equation is $\sum_{i=1}^n x_i \exp(x_i b) = \sum_{i=1}^n x_i y_i$. The mapping $b \rightarrow \sum_{i=1}^n x_i \exp(x_i b)$ is continuous and its range is $]-\infty, +\infty[$ if $\min_i x_i < 0 < \max_i x_i$, $]-\infty, 0[$ if $\max_i x_i \leq 0$ and $]0, +\infty[$ if $\min_i x_i \geq 0$; therefore, since the y_i are positive, the equation is solvable if $\min_i x_i < 0 < \max_i x_i$; otherwise a solution exists if, and only if, at least one y_i is non null. Also note that the maximum likelihood estimator $\hat{b}$ is such that $\tilde{b} = \left(\sum_{i=1}^n x_i' \cdot x_i \right)^{-1} \sum_{i=1}^n x_i' \exp(x_i \hat{b}) = \phi_n(\hat{b})$ (say), where $\tilde{b}$ is the O.L.S. estimator of b . By inverting ϕ_n , it is possible to deduce the asymptotic properties of $\hat{b}$ from those of $\tilde{b}$ (see GOURIEROUX-MONFORT (1982)).

- (3) The fact that the sample variance of y is greater than the sample mean does not invalidate the basic Poisson model. In effect, if this model is true :

$$\begin{aligned} V(y) &= V(E(y|x)) + E V(y|x) \\ &= V(E(y|x)) + E E(y|x) \geq E E(y|x) = E(y) \end{aligned}$$

- (4) When $\gamma = 0$ the covariance between $y_1(t)$ and $y_2(t)$ is equal to zero. But in that case the moment generating function of $[y_1(t), y_2(t)]$ is the product of the moment generating functions of $y_1(t)$ and $y_2(t)$. This result implies that in such models uncorrelation is equivalent to independence. In the general case γ is positive. This parameter measures the serial correlation between $y_1(t)$ and $y_2(t)$. It is positive since $y_1(t)$ and $y_2(t)$ simultaneous increase, when t increases; this positivity condition on γ does not imply a similar condition on the empirical correlation computed from the observations $y_{1i}(t)$ and $y_{2i}(t)$, when t is fixed and i is varying.

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